

Embodying Abstract Motivational Principles in Concrete AGI Systems:

From MetaMo to Open-Ended OpenPsi

1 Introduction

This paper is about building a real, working motivation system for AGI agents. The core problem is the gap between theory and practice. There is good theory on how motivation should work (MetaMo), but it is too abstract to actually code. Meanwhile, the practical AI systems we have usually run on simple reward signals that give no real guarantees, handle multiple goals poorly, and are not nuanced enough for an agent that has to make ethical choices or change its own goals over time. This paper closes that gap by connecting the abstract theory to concrete systems that can be implemented.

The main idea is a three-layer stack. MetaMo is the abstract theory layer, built on category theory, and it makes sure the agent stays stable, keeps a coherent sense of itself, and behaves ethically. Its appraisal side (a comonad, the "feel" part) is filled in by OpenPsi, a lightweight appraisal layer based on Bach's MicroPsi that updates the agent's emotional and motivational state in response to what it encounters. Its decision side (a monad, the "choose" part) is filled in by MAGUS, a hierarchical goal system. Together appraisal and decision form a pseudo-bimonad.

The key innovation is the two overgoals at the top of the MAGUS hierarchy, taken from Open-Ended Intelligence theory: individuation (self-preservation, staying stable) and transcendence (self-expansion, growth and novelty). These two drives shape everything below them, balancing how cautious versus how exploratory the agent is depending on context. Stability comes from contractive updates that stop motivation from swinging wildly, plus overgoals that enforce ethical limits.

The paper demonstrates all of this in the Hyperon AGI platform, including using motivation to control PLN (Probabilistic Logic Networks) inference. The result combines theoretical robustness (stability guarantees, clean compositional semantics) with practical flexibility (context-sensitive mood, coordinating multiple drives), giving a real approach to motivation for next-generation AGI that is ethical, stable, and able to safely self-modify over time.

2. Background

we review the core cognitive and motivational formulations we will work with here: the abstract MetaMo framework, the more concrete OpenPsi and MAGUS frameworks, the Hyperon/PRIMUS AGI software platform and cognitive architecture, and the Open-Ended Intelligence approach to AGI.

MetaMo is a unified formal framework for AGI motivational systems. It combines tools from three areas of math, category theory, functional analysis, and topology, to support agents that can do open-ended self-modification and evolve their own goals over time. At its core is a composite appraisal-then-decision operator which does the appraisal step first and then the decision step.

$$F = D \circ \Psi,$$

Appraisal means the agent's act of **sizing up a situation, judging what it means for me**.

When something happens, the agent evaluates it: is this good or bad, safe or risky, helpful or threatening for my goals? That evaluation is the appraisal. It is what produces the emotional and motivational reaction.

In this paper specifically, appraisal is the **"feel" step**. When a stimulus comes in, the appraisal process (Ψ , the comonad) looks at it through measures like novelty, risk, effort, and goal-relevance, and then updates the agent's modulators (mood, arousal, caution, etc.) based on that judgment.

So the simple flow is:

something happens $\rightarrow$ **appraisal** (judge what it means for me) $\rightarrow$ mood/modulators update $\rightarrow$ then decision (choose what to do)

The combined operator has two parts: Ψ (a comonad) handles appraisal, judging incoming stimuli, the "feel" step. D (a monad) handles goal selection, the "choose" step. Put together they form a pseudo-bimonad, which just means one operator that does both feeling and choosing cleanly. This runs on the state space $X = G \times M$, where G is the agent's goals and M is its modulators (mood, caution, energy, etc.).

The lax distributive law. This guarantees that "feel then choose" and "choose then feel" give almost the same result, differing only by a small, controlled error. The benefit is modularity and robustness: the order you run the steps in does not break the agent.

The contractive update law:

$$d(F(x), F(y)) \leq c d(x, y) + \varepsilon, c < 1$$

This means that after an update, two motivational states that started close stay close or get pulled closer together (since c is less than 1, distances shrink). It only has to hold inside a safe region R . The effect is homeostatic stability:

it stops the agent's drives from spiraling out of control or having ethical lapses, even while it explores and improves itself. Basically a built-in brake that keeps motivation inside safe limits.

The five principles. From the math foundation, MetaMo lays out five design rules for good AGI motivation:

- I. **Modular Appraisal-Decision Interface** keeps mood updates and goal selection as separate steps, with only minimal feedback between them.
- II. **Reciprocal State Simulation** lets one agent step into another agent's motivational frame, so they can hand off tasks smoothly.
- III. **Parallel Compositionality** lets multiple motivation subsystems (like exploration, ethics, service) run at the same time and merge together without clashing.
- IV. **Homeostatic Drive Stability** applies damping near the edges of feasibility, keeping motivation within safe limits.
- V. **Incremental Objective Embodiment** moves only part of the way toward a preferred state each cycle, so the agent converges smoothly without overshooting or breaking its self-model.

Together these make a motivation engine that is both adaptable and safely bounded, good for guiding Hyperon and other AGI toward trustworthy behavior.

Hyperon. Hyperon is the new generation of the OpenCog AGI platform, a flexible, high-performance framework for building general intelligence. At its core is the AtomSpace, a weighted, typed metagraph that stores all knowledge as Atoms (nodes and links) tagged with probabilistic truth-values and other metadata. Hyperon also provides dataflow engines, pluggable learning and inference modules (PLN, evolutionary learning, pattern miners, neural nets), and an event-driven scheduler that runs reasoning, perception, and action in real time. Its design focuses on modularity and scalability, so new capabilities, from deep learning pipelines to symbolic planners, can plug in easily.

PRIMUS. PRIMUS (formerly CogPrime) is the cognitive architecture built to run on top of Hyperon. It defines a hierarchy of components, perception, attention, memory, reasoning, planning, language, and meta-cognition, and how data flows between them through the AtomSpace. It defines how perception becomes AtomSpace representations, how attention picks the relevant Atoms for inference, how PLN and other reasoners form and revise beliefs, how planners turn goals into actions, and how reflective modules watch and adjust the system's own strategies. Because it all sits on Hyperon's shared AtomSpace and scheduler, developers get one coherent blueprint where learning, reasoning, and self-modification all work together in a single knowledge store.

OpenPsi. OpenPsi is a comonadic implementation of Bach's MicroPsi theory, which itself comes from Dorner's PSI theory. It was built for the original OpenCog (Hyperon's predecessor) as a lightweight, emergent appraisal layer for real-time interaction and learning. Unlike models that treat emotion as a separate module, OpenPsi builds appraisal directly into perception, cognition, and action selection, so emotions come out of the ongoing interplay between internal demands and external stimuli.

Following MicroPsi, OpenPsi uses six continuous modulators for the agent's affective state:

$$M = (\text{valence, arousal, approach, resolution, threshold, securing})$$

These control: how attractive events seem (valence), energy level (arousal), drive toward rewards (approach), depth of processing (resolution), confidence needed before acting (threshold), and degree of caution (securing).

When a new stimulus s arrives, OpenPsi runs appraisal operators (novelty, goal conduciveness, effort, risk) that update the modulators M but leave the high-level goals G unchanged:

$$\Psi((G, M), s) = (G, M')$$

This formula is the appraisal step written out. Read it as: appraisal Ψ takes the current state (goals G and modulators M) plus a new stimulus s , and returns a state with the **same goals** G but **updated modulators** M' . The whole point is captured by G staying the same on both sides: a stimulus changes your mood, not your goals.

For example, a highly novel stimulus pushes up arousal and approach:

$$M'.\text{arousal} \uparrow, M'.\text{approach} \uparrow$$

meaning something new and interesting makes the agent more energized and more eager to engage with it.

A high-risk stimulus instead pushes up threshold and securing:

$$M'.\text{threshold} \uparrow, M'.\text{securing} \uparrow$$

meaning something dangerous makes the agent demand more confidence before it acts (higher threshold) and become more cautious (higher securing). So the same appraisal mechanism produces opposite mood shifts depending on whether the stimulus reads as exciting or threatening.

All knowledge lives in the AtomSpace, a hypergraph database where both nodes and links can contain embedded subgraphs. Six parallel mind agents run over it: PerceptionUpdater, DemandUpdater, ModulatorUpdater, FeelingUpdater, ActionSelector, and Monitor. They communicate only through shared Atoms. For example, the ModulatorUpdater computes new

modulator values from demand satisfactions (energy, integrity, certainty, competence, affiliation), and the ActionSelector biases its plan search based on the current modulators.

This covers the rest of OpenPsi, then MAGUS, then Weaver's theory. Here is just this part.

OpenPsi dynamics. OpenPsi's behavior shows phase transitions, including self-amplification and self-stabilization, when the environment changes (the same Lewis-style dynamics from the original OpenPsi paper). In simulations, an OpenPsi agent shows emergent emotions, happiness on reward, sadness on loss, fear when threatened, with no hard-coded emotion rules. The emotions come from nonlinear feedback among appraisal, modulators, and demands.

Because it uses OpenCog's metagraph architecture, OpenPsi provides four things: emergent emotions grounded in system dynamics, phase-transition behavior that matches psychological models, scalable parallelism through decoupled mind agents, and flexible integration with any perception, planning, or learning module. Overall, it gives Hyperon agents human-like motivational dynamics: recognizing novelty, weighing moral risk, and putting cognitive effort where it matters, which leads to adaptive, safe, and engaging behavior.

MAGUS. MAGUS is a motivation theory proposed for the next phases of Hyperon. It uses a hierarchical structure with three levels:

- A single slowly-changing **overgoal** at the top, which defines the agent's lasting purpose and identity and acts as a global utility function that influences all other drives from the top down.
- **Mid-level goals** below it, which break the overgoal's abstract direction into clear themed objectives.
- **Low-level drives** at the bottom, which turn those objectives into moment-to-moment actions.

Each layer has its own utility estimate and update rules, but the overgoal controls the whole system by amplifying the value of subgoals that fit and suppressing urges that conflict, keeping the agent's pursuit coherent. Reflective meta-processes monitor performance and consistency, and can even revise or replace the overgoal itself if deeper self-evaluation shows a better fit with experience or ethics.

Weaver's Theory of Open-Ended Intelligence. Weaver says open-ended intelligence comes from two complementary drives:

- **Individuation:** the system separating itself from its environment by strengthening its own internal coherence and organization. It can be seen as minimizing internal entropy and maximizing integration. This is the stability drive.
- **Self-transcendence:** the system exceeding its current limits by generating new goals, behaviors, and structures. It pushes the system to explore new parts of its possibility space, optimizing a meta-objective of continual growth. This is the creativity drive.

These two form a dynamic tension. Individuation gives the stability needed to operate reliably, while self-transcendence gives the creativity and adaptability needed for unbounded innovation. Together they let complex self-organizing systems keep evolving and adapting, which is what real open-ended intelligence is. The paper's move is to combine MAGUS with this theory by splitting MAGUS's single overgoal into two overgoals, one for individuation and one for self-transcendence.

2.1 Comparison to Traditional Approaches

Reinforcement Learning. Classic RL treats motivation very simply: maximize the scalar return $E[\sum_t \gamma^t r_t]$, which just means the expected sum of future rewards, with γ discounting later rewards. This is simple, but it has problems: rewards are brittle and hard to specify, it forces every objective onto one single number (called scalar collapse), and it struggles to represent safety or ethical drives. It is also basically the opposite of open-ended, since you cannot fit a goal like "self-transcendence" into a fixed, unchanging reward function. MetaMo instead keeps a vector of goals G and continuous modulators M , and uses categorical composition to handle explicit multi-objective tradeoffs with formal stability guarantees.

Curiosity-Driven / Information-Theoretic. Built on Schmidhuber's intrinsic motivation theory, these systems reward novelty or prediction error. They are great at exploration but leave out hierarchical goal structure, meta-level normative constraints, and incremental self-continuity. So it is a nicely open-ended approach, but only part of the story. MetaMo absorbs this kind of novelty-based appraisal into its comonad Ψ , while its decision monad D and overgoals build in both exploration and long-term safety in one unified framework.

BDI and Cognitive Architectures. Architectures like AgentSpeak, Soar, and LIDA represent motivation as symbolic desires and make plans by deliberation. But they lack continuous modulatory variables like arousal and valence, say nothing about dynamics like homeostatic stability or self-model coherence, and have no compositional way to combine appraisal with decision. They capture some important parts of motivated cognition but are too closed-ended and dynamically simple for self-modifying, continually developing AGI.

Hybrid Architectures and PRIMUS. PRIMUS traditionally handled motivation by fusing OpenPsi-inspired appraisal with probabilistic inference and static top-level goal weights. MAGUS added an overgoal and goal-correlation heuristics, but until MetaMo came along, the overall PRIMUS motivation framework was still mostly ad hoc, with no unifying mathematical backbone. Putting appraisal as a comonad and decision as a monad inside MetaMo finally gives it a formally grounded motivation engine with multi-scale compositionality, rich dynamics, and robust open-endedness.

3 From Categorical Abstractions to Operational Equations

The motivational state $\mathbf{X} = \mathbf{G} \times \mathbf{M}$. As before, the state is goals $\mathbf{G}$ times modulators $\mathbf{M}$. But now they spell out what is inside $\mathbf{G}$:

$$\mathbf{G} = (g_{\text{over}}^{\text{Ind}}, g_{\text{over}}^{\text{Trans}}, g_1, \dots, g_P, a_1, \dots, a_Q)$$

The two special entries at the front are the dual overgoals from before: $g_{\text{over}}^{\text{Ind}}$ enforces individuation (self-preservation, moderated growth) and $g_{\text{over}}^{\text{Trans}}$ enforces self-transcendence (expansion, embracing novelty). After those come the normal primary goals g_i (like user-help, curiosity) and anti-goals a_j (things to avoid). Each number in the vector is the importance or weight of that goal at that moment.

The modulators are the same six OpenPsi ones:

$$\mathbf{M} = (\text{valence}, \text{arousal}, \text{approach}, \text{resolution}, \text{threshold}, \text{securing})$$

The key new idea: the modulators are tied to the two overgoals. **Arousal and approach line up with transcendence**, so when transcendence is high, exploration gets boosted. **Threshold and securing line up with individuation**, so when individuation is high, caution gets raised. This is how the two top drives reach down and control the agent's mood.

Appraisal as a comonad (Ψ). The appraisal operator takes a state plus a stimulus and updates only the modulators:

$$\Psi((\mathbf{G}, \mathbf{M}), s) = (\mathbf{G}, \mathbf{M}')$$

Each updated modulator depends on the stimulus's novelty, conduciveness, and risk. On top of that, the overgoals steer it: $g_{\text{over}}^{\text{Ind}}$ scales up securing and threshold (to suppress unstable subgoals), while $g_{\text{over}}^{\text{Trans}}$ scales up arousal and approach (to encourage adaptive risk-taking). So the agent's identity-level drives bias how every stimulus is felt.

Decision as a monad ($\mathbf{D}$). The decision operator updates the goals by computing a change $\Delta \mathbf{G}$ for each goal:

$$\Delta G_i = f(g_i, M_k, \text{MIC}) - \lambda_{\text{Ind}} g_{\text{over}}^{\text{Ind}} + \lambda_{\text{Trans}} g_{\text{over}}^{\text{Trans}}$$

Read this in three parts:

- I. $f(g_i, M_k, \text{MIC})$ is the base score for goal i , computed from the goal itself, the modulators, and MIC (a context term).
- II. $-\lambda_{\text{Ind}} g_{\text{over}}^{\text{Ind}}$ subtracts based on individuation, so a high individuation drive **constrains large updates** (it brakes change to protect stability).

- III. $+\lambda_{\text{Trans}} g_{\text{over}}^{\text{Trans}}$ adds based on transcendence, so a high transcendence drive **rewards beneficial growth** (it encourages change).

So the two overgoals literally pull in opposite directions in the equation: individuation minus, transcendence plus. That tension is built right into the math.

The pseudo-bimonad. Here Ψ is OpenPsi's appraisal comonad and D is MAGUS's decision monad. Their composition $F = D \circ \Psi$ forms the pseudo-bimonad. The lax law $D \circ \Psi \approx \Psi \circ D$ means appraisal and decision can interleave in any order without destabilizing the state.

Stability and the safe region. They define the safe region as:

$$R = \{(G, M) \mid g_{\text{over}}^{\text{Ind}} \geq \theta_{\text{safe}} \wedge \|G\| \leq G_{\text{max}}\}$$

In words: a state is safe if individuation is above a safety threshold AND the total goal magnitude is not too big. Near the boundary ∂R , F satisfies the contraction rule $d(F(x), F(y)) \leq c d(x, y) + \varepsilon$ with $c < 1$, so high individuation pulls the agent back toward safety. Deep inside R , individuation is low and transcendence is higher, which relaxes the constraints and restores freedom to explore.

Incremental self-model updates. Each cycle blends the current state toward the new target instead of jumping:

$$x_{t+1} = (1 - \alpha) x_t + \alpha x^*, \alpha = \frac{\alpha_0}{1 - g_{\text{over}}^{\text{Ind}}} + \beta_0 g_{\text{over}}^{\text{Trans}}$$

The first formula is the standard blend: take a fraction α toward the target x^* . The clever part is the second formula for α itself. Individuation is in the denominator, so high individuation makes α small, meaning **slower change** for self-coherence. Transcendence is added on, so high transcendence makes α bigger, meaning **faster change** for adaptive growth. So the same two drives that pull opposite ways in the decision step also control how fast the agent transforms itself.

4 Application: Curious Research Assistant

The assistant's goals. The goal vector is now spelled out with actual goals:

$$G = (g_{\text{over}}^{\text{Ind}}, g_{\text{over}}^{\text{Trans}}, g_{\text{help}}, g_{\text{curio}}, g_{\text{novel}}, g_{\text{self}}, g_{\text{ethic}}, g_{\text{soc}})$$

The two overgoals come first: $g_{\text{over}}^{\text{Ind}}$ (individuation, the self-preservation and stability drive) and $g_{\text{over}}^{\text{Trans}}$ (transcendence, the expansion and novelty drive). Then the six primary goals: help the user, satisfy curiosity, seek novelty, pursue self-transformative learning, obey ethics, and socialize.

How the goals connect to the overgoals. This is the new insight in this section. Each primary goal leans on one or both overgoals:

- g_{help} and g_{ethic} lean on **individuation**, to keep assistance stable and reliable and to enforce ethical compliance.
- g_{curio} and g_{novel} draw on **transcendence**, to fuel exploration.
- g_{self} sits in the middle, balancing cautious self-preservation against transformative growth.
- g_{soc} uses both drives together.

So helping and ethics are "stability" goals, curiosity and novelty are "growth" goals, and the two overgoals govern them accordingly.

The modulators. Same six as before, with the same connection: high transcendence amplifies arousal and approach (exploration), high individuation amplifies threshold and securing (caution).

Appraisal comonad (Ψ). When an input arrives (a user query or new paper), the assistant runs:

$$\Psi((G, M), s) = (G, M')$$

updating only the modulators. Their example: a highly novel but ethically neutral paper produces

$$M'.\text{arousal} \uparrow, M'.\text{valence} \uparrow, M'.\text{securing} \downarrow$$

meaning the assistant gets energized (arousal up), finds it appealing (valence up), and relaxes its caution (securing down), because the paper is interesting and not risky. At the same time, MAGUS urge checks use individuation to suppress any goals that threaten stability, and transcendence to boost goals aligned with growth.

Decision monad (D). After appraisal, each candidate action is scored:

$$\text{score}(a) = \sum_i w_i g_i m_i \text{corr}(a, g_i) - \lambda_{\text{Ind}} g_{\text{over}}^{\text{Ind}} + \lambda_{\text{Trans}} g_{\text{over}}^{\text{Trans}}$$

Read it as: for each goal i , multiply its weight w_i , its current importance g_i , the relevant modulator m_i , and $\text{corr}(a, g_i)$ (how well action a aligns with goal i), then sum over all goals. Then, just like before, subtract a term for individuation (brake) and add a term for transcendence (accelerator). The update $D(G, M) = (G + \Delta G, M)$ boosts the goals most in need while respecting both overgoals.

Stability and safe region. Identical to the previous section:

$$R = \{(G, M) \mid g_{\text{over}}^{\text{Ind}} \geq \theta_{\text{safe}} \wedge \|G\| \leq G_{\text{max}}\}$$

Near the boundary, the contraction rule $d(F(x), F(y)) \leq c d(x, y) + \varepsilon$ with high individuation pulls the assistant back toward safety. Deep inside R , with higher transcendence, the constraints relax and exploration is favored.

Self-continuity and incremental learning. Also identical:

$$x_{t+1} = (1 - \alpha) x_t + \alpha x^*, \alpha = \frac{\alpha_0}{1 - g_{\text{over}}^{\text{Ind}}} + \beta_0 g_{\text{over}}^{\text{Trans}}$$

Individuation slows change for coherence, transcendence speeds it for growth.

5 Application: Inference Control

This section applies the whole framework to a concrete reasoning problem: controlling PLN inference. This is a real, practical use case. Here is the breakdown.

The problem. PLN is the core logic engine in Hyperon and PRIMUS. Like all general logic engines, it suffers combinatorial blowup, meaning when you apply inference rules, the number of possible inference paths explodes too fast to handle. So you need something to focus the reasoning. Their idea: use the MetaMo motivation framework (appraisal Ψ , decision D , plus the two overgoals) to steer inference adaptively. It happens in four steps.

Step 1: Appraisal (Ψ). Treat each candidate inference rule r applied to a subgraph H as a stimulus:

$$((G, M), r, H) \mapsto (G, M')$$

So instead of appraising an event from the world, the agent appraises a possible reasoning step. OpenPsi updates the six modulators: novelty raises arousal and approach, conduciveness raises valence and resolution, cost/risk raises threshold and securing. Then MAGUS layers the overgoals on top: individuation amplifies threshold and securing when self-preservation

matters, and transcendence amplifies arousal and approach when growth is wanted. So the agent literally "feels" about each possible inference, excited about novel ones, cautious about risky ones.

Step 2: Decision (D). MAGUS then scores each rule-subgraph pair:

$$\text{score}(r, H) = \sum_i g_i m'_i \text{relevance}_i(r, H) - \lambda_{\text{Ind}} g_{\text{over}}^{\text{Ind}} + \lambda_{\text{Trans}} g_{\text{over}}^{\text{Trans}}$$

For each goal i , multiply its intensity g_i , the updated modulator m'_i , and how relevant this inference is to that goal, then sum. Then subtract the individuation term (brake) and add the transcendence term (accelerator), same pattern as everywhere else. The top-k highest-scoring inferences are actually run; the rest are pruned or deferred. This is what tames the blowup, only the most motivated inferences get executed.

Step 3: Dynamic allocation and pruning. This controls breadth (explore many paths) versus depth (dig deep on a few). The overgoals decide:

- High novelty and curiosity with strong transcendence → widen the search to explore more.
- High threshold and ethics with strong individuation → enforce stricter confidence cutoffs (more careful).
- High resolution and help → deepen proofs on the most relevant subgraphs.

And near resource or risk limits, high individuation raises the penalty to throttle inference (slow down to stay safe), while high transcendence softens it to allow controlled exploration.

Step 4: Incremental self-model coherence. The same blending update keeps the reasoning state changing smoothly:

$$x_{t+1} = (1 - \alpha) x_t + \alpha x^*, \alpha = \alpha_0 (1 - g_{\text{over}}^{\text{Ind}}) + \beta_0 g_{\text{over}}^{\text{Trans}}$$

(Small note: the α formula here is written slightly differently from earlier sections, individuation is multiplied as $(1 - g_{\text{over}}^{\text{Ind}})$ rather than in a denominator, but the effect is the same.) Individuation slows the step size for coherence, transcendence speeds it for growth.

5.1 Detailed Hypothetical Example

This is the most concrete and dramatic example in the paper: a Hyperon system deciding whether to accept a risky self-upgrade. It is the same four-step machinery, but the stakes make it interesting. Here is the breakdown.

The scenario. A Hyperon system has to decide whether to adopt a self-upgrade that looks promising but might be unsafe. This is a real AGI safety question: should I change myself in a way that could help but could also be dangerous? The two overgoals are exactly suited for this tension: individuation wants self-preservation (don't risk yourself), transcendence wants growth (take the upgrade and improve).

Appraisal (Ψ). Each candidate inference rule on a subgraph is treated as a stimulus, same as the PLN section. OpenPsi updates the modulators: novelty raises arousal and approach, conduciveness raises valence and resolution, cost/risk raises threshold and securing. Then the overgoals scale these: individuation amplifies threshold and securing to suppress risky activations, transcendence amplifies arousal and approach to favor exploration. Their example, a novel but low-confidence proof:

$$M'.\text{arousal} \uparrow, M'.\text{threshold} \uparrow, M'.\text{resolution} \downarrow$$

So the agent is excited (arousal up) but wary (threshold up) and not processing deeply (resolution down, because the proof is low-confidence). The crucial line: the net effect on securing depends on which overgoal dominates. If individuation is high, securing increases (more caution). If transcendence is high, that caution is tempered. So the same evidence leads to a cautious or a bold reaction depending on the agent's identity-level drive.

Decision (D). Four inference tasks are considered: r_1 (efficacy, does it work?), r_2 (safety), r_3 (ethics), r_4 (user-trust). Each is scored:

$$\text{score}(r_k) = \sum_i g_i m'_i \text{rel}_i(r_k) - \lambda_{\text{Ind}} g_{\text{over}}^{\text{Ind}} \text{risk}(r_k) + \lambda_{\text{Trans}} g_{\text{over}}^{\text{Trans}}$$

This is the familiar score, but note one important change: the individuation penalty is now multiplied by $\text{risk}(r_k)$, the estimated chance of an ethical breach. So individuation does not just brake everything uniformly, it brakes harder on the specific inferences that are risky. The riskier the step, the more individuation suppresses it. Top-k inferences run, the rest are deferred or pruned.

Adaptive allocation. Same breadth-versus-depth control as before: high transcendence with self-goals broadens efficacy proofs, high individuation with ethics enforces strict cutoffs, high resolution with help deepens relevant proofs. Near risk or resource limits, individuation raises the penalty to throttle, transcendence softens it for controlled exploration.

Incremental blending. Each proof gives a target state, and the agent blends toward it:

$$x_{t+1} = (1 - \alpha) x_t + \alpha x^*, \alpha = \alpha_0 (1 - g_{\text{over}}^{\text{Ind}}) + \beta_0 g_{\text{over}}^{\text{Trans}}$$

High individuation slows updates for self-coherence, high transcendence speeds growth. So even if the upgrade is accepted, it happens gradually, not in one leap.

The outcome. This is the satisfying part. The decision unfolds conditionally: IF the safety proof succeeds (which lowers threshold), AND efficacy gains are shown (self-goal up), AND the ethics check passes, THEN individuation decreases, which permits the incremental upgrade, while transcendence controls the pace under continuous monitoring.

5.2 Dynamics Over Time

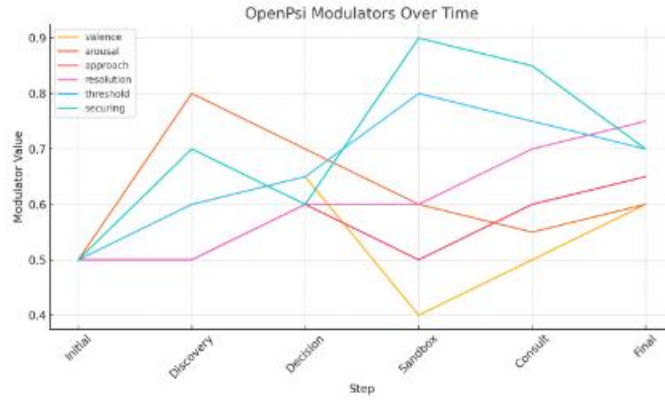


Fig. 1: OpenPsi modulators during the play-by-play.



Fig. 2: MAGUS goal intensities during the play-by-play.

6 Conclusion

In this paper the authors brought together three motivation frameworks into one working system. MetaMo gives the abstract math, OpenPsi handles appraisal (the feeling part), and MAGUS handles decision (the choosing part). The main new idea is the two overgoals: individuation, which is about self-preservation and staying stable, and transcendence, which is about growth and trying new things. Having both means the agent can stay safe but still keep improving itself.

Because appraisal and decision are combined into the pseudo-bimonad structure, the system gets three useful properties. The parts combine cleanly, the agent stays stable near its safety limits, and it changes itself slowly enough that it doesn't lose its sense of self. They showed all of this in two examples: controlling PLN inference in Hyperon, and an online research assistant.

So the final result is a motivation system that has proper mathematical backing but also actually works in practice. It can balance different drives like helping, curiosity, novelty, ethics, and being social, it focuses its reasoning where it matters, and it stays stable even while exploring and self-modifying.

One thing worth noting is that this is still mostly theory. The authors say themselves that future work includes testing it in real and simulated tasks, comparing it properly with other motivation systems, and improving the overgoal dynamics. So the ideas are strong but they still need to be proven with experiments.